DIPLOMA PROGRAMME PRACTICE · SL · SET B

Biology / Paper 2

Question booklet · 50 marks · 90 minutes

Show working, units and suitable precision. Use a calculator. Use separate answer paper for additional working and diagrams. Answer all of Section A and one of two Section B questions. Each extended response carries 15 content marks and one mark for overall quality.

SECTION A · COMPULSORY

QUESTION 1 [12]

Inferring grazing effects in a rocky pool

At each of ten comparable pools, researchers randomly assigned two equal rock plots to either a grazer-exclusion cage or an open-sided cage that admitted grazers. Both treatments used the same mesh above the rock. The table gives mean algal dry mass. Values are g m^{-2}.

Time	Exclusion	Open-sided
Start	18	19
Week 2	27	22
Week 4	42	24

(a) Calculate the difference between treatments in change from start to week 4, and give its units. **[3]**

(b) Explain why the open-sided treatment is more informative than simply comparing a caged plot with a bare rock. **[2]**

(c) Explain why the pool is a sensible unit for pairing plots and identify the independent replicate count. **[3]**

(d) State a conclusion supported by the means and one measurement needed to assess consistency among pools. **[2]**

Paper 2 · Question 1 (continued)

(e) Suggest a second mechanism besides grazing that could differ between cage types and a way to test it. **[2]**

SECTION A · COMPULSORY
QUESTION 2 **[8]**

Two plant glucose polymers

A plant stores glucose in polymer S and uses polymer W to reinforce cell walls. Hydrolysis of purified samples yields glucose in both cases.

(a) Identify S and W and distinguish their glucose building blocks. **[3]**

(b) Connect the arrangement of cellulose chains to wall strength. **[3]**

(c) Explain why storing glucose as starch helps maintain cellular water balance. **[2]**

SECTION A · COMPULSORY
QUESTION 3 [7]

Interpreting vaccination outcomes

In a randomized trial with complete follow-up, 24 of 800 vaccinated participants and 64 of 800 comparison participants developed laboratory-confirmed disease. Both groups received identical surveillance.

(a) Calculate risk in each group and the relative risk reduction. **[3]**

(b) Distinguish that reduction from the absolute change in risk. State why random assignment matters. **[2]**

(c) Explain how vaccination can reduce disease despite some vaccinated participants still developing it. **[2]**

SECTION A · COMPULSORY
QUESTION 4 [7]

Reading a sequence of cell images

The time series shows one animal cell in culture. Times are measured from a common starting point. The images do not show individual chromosomes clearly.

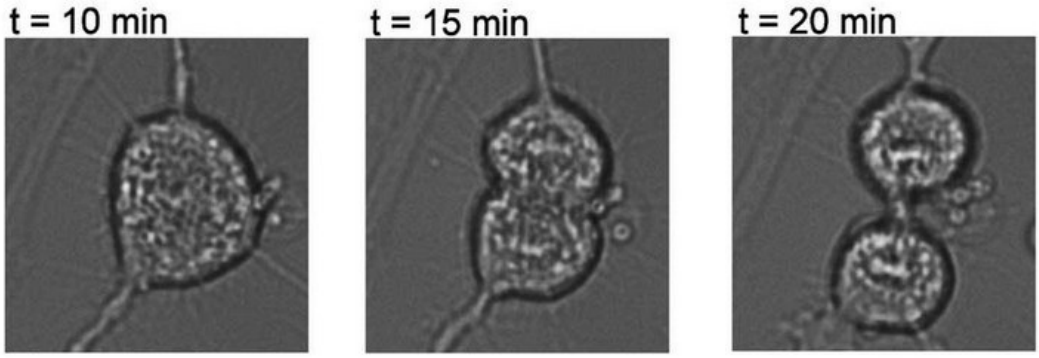


Image source: A.B. Jonsdottir et al. Cellular Oncology 31 (2009) 191–201 CC BY 2.0

Time series of a cultured cell. A. B. Jonsdottir and colleagues, Cellular Oncology 31 (2009), 191–201, CC BY 2.0, as credited in the supplied paper.

(a) Describe two changes visible between the first and last images. **[2]**

(b) Identify the division process directly supported by the shape change and distinguish it from division of a nucleus. **[2]**

(c) A student claims the images prove the daughter cells have identical DNA sequences. Evaluate the claim and propose evidence. **[3]**

SECTION B · EXTENDED RESPONSE
QUESTION 5 [16]

Supplying an exercising muscle

A runner begins sustained exercise. Consider the path from air to a working muscle cell and how that cell uses the delivered oxygen.

(a) Outline the main mechanical changes during inspiration in a human at rest. **[3]**

(b) Explain how the alveolar exchange surface allows rapid transfer of oxygen into blood. **[5]**

(c) Explain how increased oxygen delivery supports sustained movement, linking transport to energy transfer. **[7]**

[1] Overall quality of the extended response.

SECTION B · EXTENDED RESPONSE
QUESTION 6 [16]

Variation, selection and conservation

An isolated lizard population varies in a heritable trait affecting survival on hot rock. A conservation scheme will establish a second population.

(a) Outline sources of genetic variation in sexually reproducing organisms. **[3]**

(b) Explain how directional selection could change the trait distribution over many generations. **[5]**

(c) Discuss choices that would help the second population retain diversity and allow the conservation team to assess its success. **[7]**

[1] Overall quality of the extended response.